

teaching method

hamed obeidat

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In a study, the teacher determined whether a new teaching method improved student performance. A random sample of students was drawn to study the difference before and after the method was implemented.

Table 1: Table1:student grade before and after

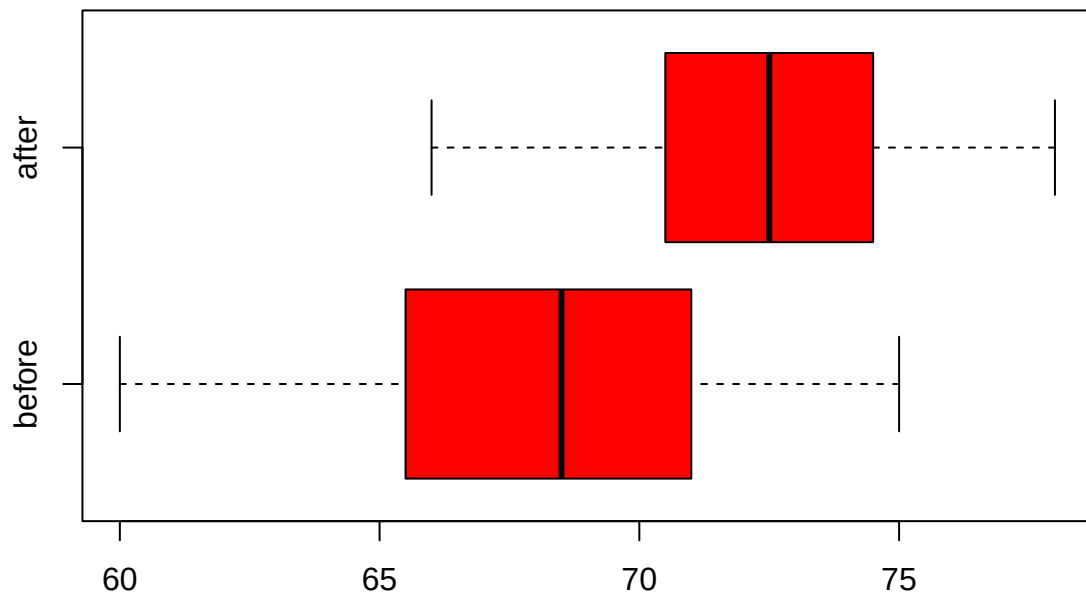
before	after
65	72
70	75
68	70
60	66
75	78
72	74
66	71
69	73

Show table is number of student before and after .

Descriptive statistic:

##	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
## before	1	8	68.12	4.58	68.5	68.12	4.45	60	75	15	-0.24	-1.03	1.62
## after	2	8	72.38	3.58	72.5	72.38	2.97	66	78	12	-0.20	-0.92	1.27

The average student grades before using the method were 68.12 ,The average score when using the method was 72.38 ,The highest score was achieved before using the method is 75 ,The less score was achieved before using the method is 60,The highest score was achieved after using the method is 78 ,The less score was achieved after using the method is 66 .



The graph shows that the mean grade using the after are higher than the before and , while the spread is fairly similar.

Inference statistic:

Check the assumptions :

```
## Loading required package: ggplot2
```

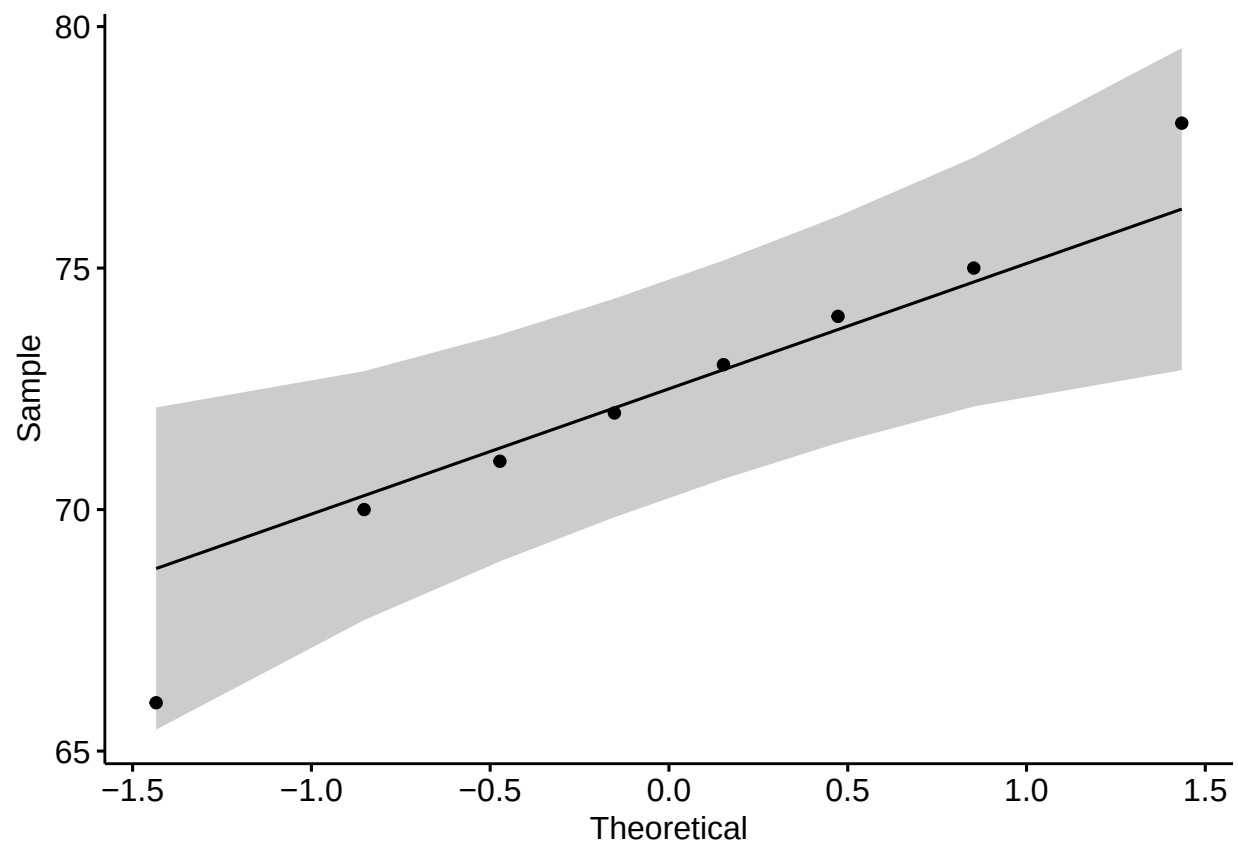
```
##
```

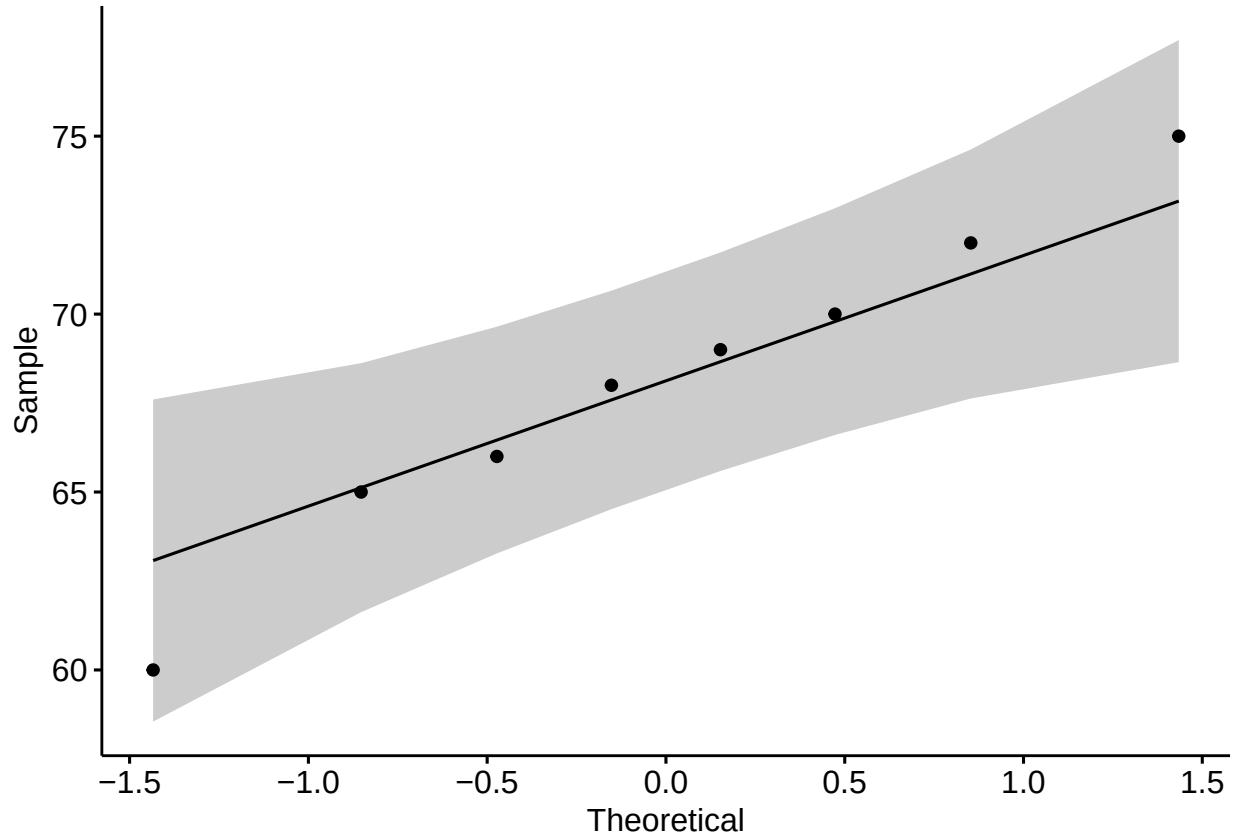
```
## Attaching package: 'ggplot2'
```

```
## The following objects are masked from 'package:psych':
```

```
##
```

```
## %+%, alpha
```





The graph shows that the grades are of a normal distribution.

```
## [1] -7 -5 -2 -6 -3 -2 -5 -4
```

```
## [1] 3.357143
```

the two sample equal variance

H_0 : There is no statistically significant difference two teaching method H_1 : There is statistically significant difference two teaching method

Table 2: Result paired sample t test

				before	after	95% confidence
test	t	df	p-value	mean	mean	interval
t paired sample t test	-6.5607	7	0.0001578	-4.25	NA	(-Inf , -3.02)

The results showed that there were statistically significant differences, where the value reached p-value=7 0.0001578 less than 0.05 ,DF=7 ,This means that the new method improved students' grades.